

A Lab Project Report

On

Diameter Measurement of the Object in image

Submitted in partial fulfilment for award of the degree of Bachelor of Technology in

Department of electronic and communication engineering

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Vision of Institution – BRECW

- BRECW develops confident and articulate young women into dynamic Engineers equipped with skills, knowledge, values and an attitude to contribute to the society.

Mission of Institution- BRECW

- BRECW is committed to providing a challenging, enriching, safe and supportive technical learning environment through its core values of responsibility, respect and compassion.
- Fosters intellectual, spiritual and personal development of young women so that they develop the tools necessary to lead meaningful lives.
- Offers academic curriculum along with an extensive co-curricular program with the support of dedicated staff who ensure that students identify their strengths and develop their skills such as teamwork, leadership, creativity and entrepreneurship
- Develops independent, adaptable thinkers with a passion for learning, courage to take risks and initiative to apply what is learned.

Department of Electronic and Communication Engineering

Vision

- ECE department envisions developing technically competent and meritorious women engineers with a keen sense of social responsibility.

Mission

- To provide a challenging and value-based education, enriching knowledge of young engineers in the field of Electronics and Communication Engineering.
- To strive for the intellectual and personal development of young women to build a healthy society by improving the quality of life through the application of Electronics. •To inculcate self-confidence, teamwork, leadership, and entrepreneurship in students through curricular, co-curricular and extracurricular activities.
- To develop adaptable thinking and the ability to apply the techniques of communication Innovatively in a realistic environment for the current and future technological requirements

Program Educational Objectives of ECE Department (PEO'S)

Upon completion of the program, electronic engineers will be able to:

PEO1: Solve complex problems by using their expertise in analyzing and developing potential models using modern scientific tools.

PEO2: Prioritize their professional development through interpersonal, leadership, and social skills, catering to the needs of society with ethics and integrity.

PEO3: Exhibit sustained learning adapting to changing professional needs.

Program Specific Outcomes of ECE Department (PSO'S)

PSO1:Able to design,develop and analyses systems in the field of Electronics, Communications & Networking, Signal & Image processing,VLSI technology and Embedded systems.

PSO2: Demonstrate expertise in the use of software and hardware required in real-life applications.

Department of Electronic and Communication Engineering

Program Outcomes (POS)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
12. **Life-long learning:** Recognize the need for and have the preparation and ability to engage in

DIAMETER MEASUREMENT OF OBJECT

- **AIM** : Diameter measurement of object
- **APPARATUS** : Software: MATLAB
Hardware:Laptop/PC
- **THEORY** :

This project presents a Student Grade Evaluation System developed using MATLAB as an application of basic Signals and Systems concepts. The program processes student marks as input data and categorizes them into different grade levels using conditional logic, similar to signal classification in system analysis. With a predefined grading scale, the system assigns grades automatically without any user interaction during execution, ensuring accuracy and consistency. This project highlights the practical use of mathematical logic and signal-based decision processing in MATLAB, making it a useful tool for understanding real-time system behavior in academic evaluation scenarios. Additionally, it demonstrates how computation and decision-based systems can be modeled efficiently using MATLAB. Overall, this project bridges theoretical signal processing concepts with a simple real-world educational application.

- **objective**

- To design a MATLAB-based system for student marks evaluation.
- To calculate total marks based on the performance and percentage automatically.
- To grade student performance using conditional statements.
- To display formatted student result output.
- To develop a simple automated result-processing systems for students
- To reduce manual calculation errors in student evaluations.
- To practice basic arithmetic operations in MATLAB.
- To implement decision making structure based on percentage ranges.
- To create a program that can be extended to multiple students in the future. To understand logical flow in program design and execution.
- To improve beginner friendly programming skills in MATLAB.

EXPLANATION OF CODE:

Import Image

Open the MATLAB software and in the application section; download the Image Processing Tool Box. Create a new MATLAB script file. Refer to Figure 2 to begin adding code to import the desired image to measure into the MATLAB workspace. The first few lines clear the workspace to remove any previous variables and clear the command window. It is important that the Current Folder that you are working out of be the folder that contains both the script file and image. The command `imread` reads an image and converts it into a “3-dimensional” matrix in the RGB color space. The image used in this tutorial is `ball.jpg` (Figure 1), which is a 534 by 401 pixel image. The `imread` function converts this into a matrix that is 401x534x3 (Rows x Columns x RGB). The final dimension (RGB) corresponds to a red, green and blue intensity level. Use `imshow` to view the produced image in a new window



Figure 1; Original Image, ball.jpg

```
AN_Tutorial.m  < +
1  %% Import Image
2
3  clear;
4  clc;
5
6  obj = imread('ball.jpg');
7  imshow(obj);
8
```

Figure 2; Code to Import an Image

Workspace	
Name	Value
obj	401x534x3 uint8

Figure 3; obj variable in workspace

Segment Image

Follow the code in Figure 4 below to segment the image into a binary image to differentiate the background from the desired objects. The first step taken is to divide the image into three images based on the intensities of each red, green and blue component within the image. This is Color Based Image Segmentation. You can see from Figure 5 that the blue plane is the best choice to use for Image Thresholding because it provides the most contrast between the desired object (foreground) and the background. Image Thresholding takes an intensity image

and converts it into a binary image based on the level desired (See line 25). A value between 0 and 1 determines which pixels (based on their value) will be set to a 1 (white) or 0 (black)). To choose the best value suited for your application right-click on the value and at the top of the menu and select “Increment Value and Run Section”. Set the increment value to 0.01 and choose the best value at which to threshold. Figure 5 shows the result of the Image Thresholding at 0.37. You can see that the image (Top-right of Figure 6) has been segmented

```
9  %% Segment Image
10
11  %Divide image "obj" into its respective RGB intensities
12  red = obj(:,:,1);
13  green = obj(:,:,2);
14  blue = obj(:,:,3);
15
16  figure(1)
17  subplot(2,2,1); imshow(obj); title('Original Image');
18  subplot(2,2,2); imshow(red); title('Red Plane');
19  subplot(2,2,3); imshow(green); title('Green Plane');
20  subplot(2,2,4); imshow(blue); title('Blue Plane');
21
22  %Threshold the blue plane
23  figure(2)
24  level = 0.37;
25  bw2 = im2bw(blue,level);
26  subplot(2,2,1); imshow(bw2); title('Blue plane thresholded');
27
```

Figure 4; Code to Segment the Image

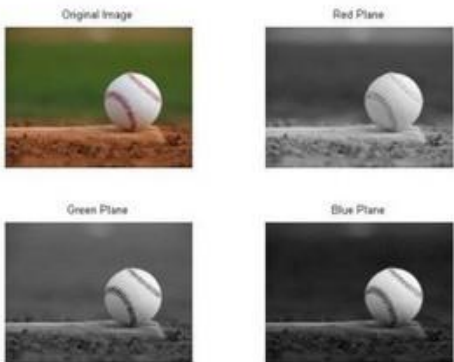


Figure 5; Color Thresholding

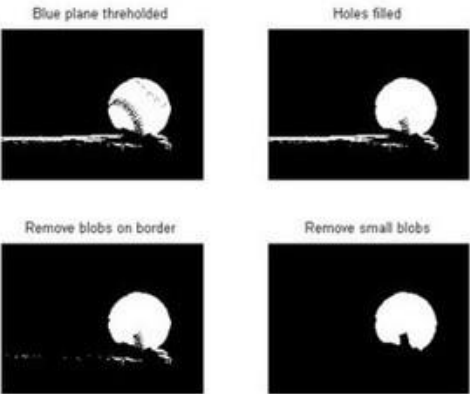


Figure 6; Complete Segmentation and Cleanup Image

Segmentation Continued (Remove Noise)

As you can see from the top-left image in Figure 6 there is quite a bit of “noise” and we need to clean the image up significantly to improve the accuracy of our diameter measurement. Refer to Figures 6 & 7 on the procedures taken to clean up the image and provide a more uniform blob to analyze. Blobs in this document are any collection of white pixels that touch to create a cohesive and distinct object.

```
28 %% Remove Noise
29
30 %Fill any holes
31 fill = imfill(bw2,'holes');
32 subplot(2,2,2); imshow(fill); title('Holes filled');
33
34 %Remove any blobs on the border of the image
35 clear = imclearborder(fill);
36 subplot(2,2,3); imshow(clear); title('Remove blobs on border');
37
38 %Remove blobs that are smaller than 7 pixels across
39 se = strel('disk',7);
40 open = imopen(fill,se);
41 subplot(2,2,4); imshow(open); title('Remove small blobs');
42
```

Figure 7; Code to Cleanup Image/Remove Noise

Measuring Image

The image in the bottom-right corner of Figure 6 is the result of all image segmentation and cleanup procedures to provide one distinct and cohesive blob, which represents the ball in the original image. Having the original image in a binary form such as this will make it easy for other functions built into MATLAB to quickly analyze the region and a host of different information. The regionprops function is the tool that will provide the MajorAxisLength of the blob in the image. As you can see, by not suppressing line 45 (Figure 8) with a semi-colon, the diameter is displayed in the Command Window (Figure 10).

```
43 %% Measure Object Diameter
44
45 diameter = regionprops(open,'MajorAxisLength')
46
47 %Show result
48 figure(3)
49 imshow(obj)
50 d = imdistline; %Include a line to physically measure the ball
51
```

Figure 8; Code to Measure the Object Diameter

Output:

The diameter is now displayed in the Command Window to be approx. 170 pixels across. This was verified in Figure 9 by using the imdistline function in line 50 (Figure 8). As you can see between the two figures, the value calculated by the code was very close to the manual measurement in Figure



Figure 9; Original Image Manually Measure



Figure 10; Diameter Output in the Command Window